

Ground Spin Strike: Presentation of Impact Sensations by a Floor-Mounted Turntable

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Figure 1: Ground Spin Strike: a floor-mounted turntable that enhances impact sensation.

Abstract

This study presents a floor-mounted turntable designed to enhance impact sensation in virtual reality (VR). The idea is to mechanically reproduce the reaction force from the floor that is generated on the soles of the feet during striking motions. Previous studies on impact rendering have mainly employed wearable or handheld devices to deliver feedback to the hands, often requiring users to hold heavy hardware. On the other hand, the reaction force generated during striking propagates from the hands through the body and is eventually exerted at the interface between the feet and the ground. Leveraging this physical property, we aim to enhance impact sensation by delivering force feedback to the feet. We demonstrate a VR-based sports application using the proposed floor-mounted turntable. Users stand on the device, and during striking actions such as swinging a racket or punching in a boxing game, the turntable rotates instantaneously to simulate the reaction force, thereby enhancing the realism of the impact experience.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; **Haptic devices**.

Keywords

Impact Sensation, Haptics, Ground-Reaction Force, Floor Interface, Virtual Reality

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1 Introduction

When designing experiences involving striking motions in VR environments, providing physical impact sensations in addition to visual information enhances the quality of the experience [Weng et al. 2025]. Such impact sensations during striking consist of vibrations (high-frequency components) and reaction forces (low-frequency components) [Shi and Shen 2024]. Previous studies have primarily adopted approaches that provide force feedback to the hands using wearable or handheld devices [Lee and Choi 2024; Tsai et al. 2022]. However, because both wearable and handheld devices are ungrounded (i.e., they are supported only by the user's body), it is difficult to generate sufficient external reaction forces [Fan et al. 2024].

In this study, we leverage the fact that the forces generated during striking propagate through the body and are eventually exerted at the interface between the feet and the ground. Based on this mechanical principle, we propose a method to simulate reaction forces



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at the feet using a floor-mounted turntable that rotates rapidly upon impact. This approach induces whole-body perception of reaction forces originating from the feet without relying on hand-based stimulation, thereby enhancing impact sensation. Furthermore, because the proposed method does not require force-feedback devices for the hands, it enables substantial physical feedback with a simple system configuration while preserving users' freedom of movement.

2 Method

When a user performs a striking motion by swinging the right hand forward to make contact, a rotational moment is exerted on the body by the impact target, resulting in an external force that rotates the entire body in a clockwise direction (Figure 2(a)). Simultaneously, friction with the floor prevents the body from slipping, generating a counteracting counterclockwise reaction force at the foot-ground interface. In other words, the reaction force generated during striking is intrinsically linked to the rotational forces occurring at the feet.

Based on this mechanical relationship, we hypothesized that applying counterclockwise angular acceleration to the feet would be effective for simulating the reaction forces during striking. However, continuous unidirectional rotation causes the user's orientation to drift from the initial position. Therefore, a control mechanism is required to promptly return the turntable to its initial orientation after the stimulation.

Accordingly, our method adopts a two-phase control scheme (CCW/CW scheme), in which the turntable is first accelerated counterclockwise and then immediately rotated clockwise. This strategy exploits the human tendency to perceive sensory input more strongly during the initial phase of stimulation, thereby emphasizing the initial counterclockwise acceleration component.

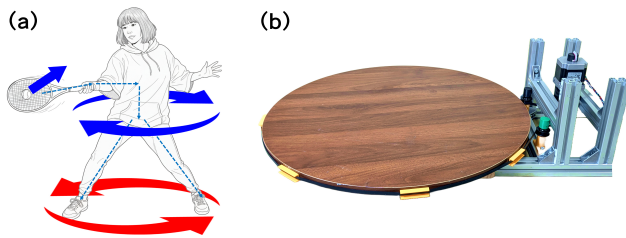


Figure 2: (a) Direction of force applied to the body during impact, (b) Turntable device for rendering impact sensation.

3 Hardware

To validate the feasibility of the proposed method, we developed a prototype capable of precisely controlling the rotation of a turntable (Figure 2(b)). The system comprises a stepper motor (NEMA23, Phidgets), a timing belt transmission mechanism, and a turntable with a diameter of 60 cm. An encoder attached to the motor shaft measures the rotation angle, facilitating stable and accurate closed-loop position control.

Impact sensations are rendered through a sequential motion in which the turntable rotates first in one direction and then immediately returns to the opposite direction. Specifically, a rotational

command with a duration of approximately 0.1 s is issued in synchronization with the detected striking input, followed by a control sequence that restores the turntable to its initial angular position. The rotational parameters were empirically tuned to ensure that the induced reaction forces are clearly perceived as impact sensations without causing excessive postural displacement or instability.

Performance measurements indicated that with a 65 kg participant standing on the device, the maximum rotation angle was approximately 4°, and the latency from command input to the onset of rotation was less than 0.1 s. Our preliminary test with seven participants also confirmed that our two-phase control scheme effectively enhanced the impact experience.

4 Demo Experience

Users stand on a floor-mounted turntable, equipped with a head-mounted display, and VR controllers in both hands. In the virtual environment, they experience interactive sports scenarios such as tennis and boxing. When users perform striking actions, such as swings or punches, and make contact with virtual objects, the turntable beneath their feet rotates momentarily, rendering impact sensations to the entire body. Our preliminary investigations indicate that while the sensation of whole-bodily rotation is not clearly perceived, the impact sensation associated with reaction forces is distinctly perceived.

In this demonstration, users can experience differences in impact sensation across various feedback modalities by adjusting the gain of rotational feedback from the turntable, as well as toggling the vibration feedback from the VR controllers. Each session is conducted individually, lasting several minutes. For safety, sufficient clearance is maintained around the turntable to prevent collisions with surrounding objects in the event of a loss of balance, and a handrail is installed to assist users in maintaining stability. In addition, staff members are constantly present to monitor the experience.

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